ChE 341 – Chemical Process Control –Spring 2020
Student Learning Outcome 1

Rubric for assessment problem on final exam (Problems #1,2,3,4)

1. **Design of PID Feedback Controller for Tank Level Control.**
2. **High Order Systems, Skogestad's Approximations and Temperature Control**
3. **Ratio Controller for a Waste Treatment Facility**
4. **Time Delay Compensation with Smith Predictor**

SLO 1 Performance Indicator (PI)	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject **Problem 2A-2F** **100 pts**	Does not understand the connection between mathematical models and the system or process to be analyzed or designed **(<40 pts)**	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development **(40-60 pts)**	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering **(>60 pts)**
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra **Problem 1A,1B,1C,1D** **100 pts**	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems **(<40 pts)**	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving **(40-60 pts)**	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems **(>60 pts)**
4. Translates academic theory into engineering applications **Problem 4A,4B** **100 pts**	Does not appear to grasp the connection between theory and the problem **(<40 pts)**	Some gaps in understanding the application of theory to the problem and expects theory to predict reality **(40-60 pts)**	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality **(>60 pts)**
7. Solutions Creativity Alternatives **Problem 3A,3B,3C,3D** **100 pts**	Demonstrates solutions implementing simple applications of one formula or equation with close analogies to class/lecture problems **(<40 pts)**	Demonstrates solution with integration of diverse concepts or derivation of useful relationships involving ideas covered in course concepts; however, no alternative solutions are generated **(40-60 pts)**	Demonstrates creative synthesis of solution and creates new alternatives by combining knowledge and information **(>60 pts)**

RESULTS BY PERFORMANCE INDICATOR				
Rubric #	**1**	**2**	**4**	**7**
Mapped SO	1	1	1	1
Average	2.44	2.67	1.56	2.78
Std Deviation	0.92	0.59	0.92	0.55
S	40	40	16	48
D	3	11	4	3
U	9	1	32	1
Checksum	**52**	**52**	**52**	**52**

RESULTS BY STUDENT OUTCOME			
SO	**1**		
Average Score	2.36		
Standard Dev.	0.48		

Average scores were fairly uniform except for No. 4 Performance Indicator, which is an indication on how students apply new ideas using their background obtained from the lectures. The low average may also be explained by the fact that the most critical lectures on process control were delivered on line due to Covid-19 and some students might have had hard time grasping advanced topics of the subject. A large number of students were strong in solving engineering problems using theoretical process control concepts , and concepts of algebra, calculus, differential equations and/or linear algebra. Overall performance was satisfactory for all performance indicators.

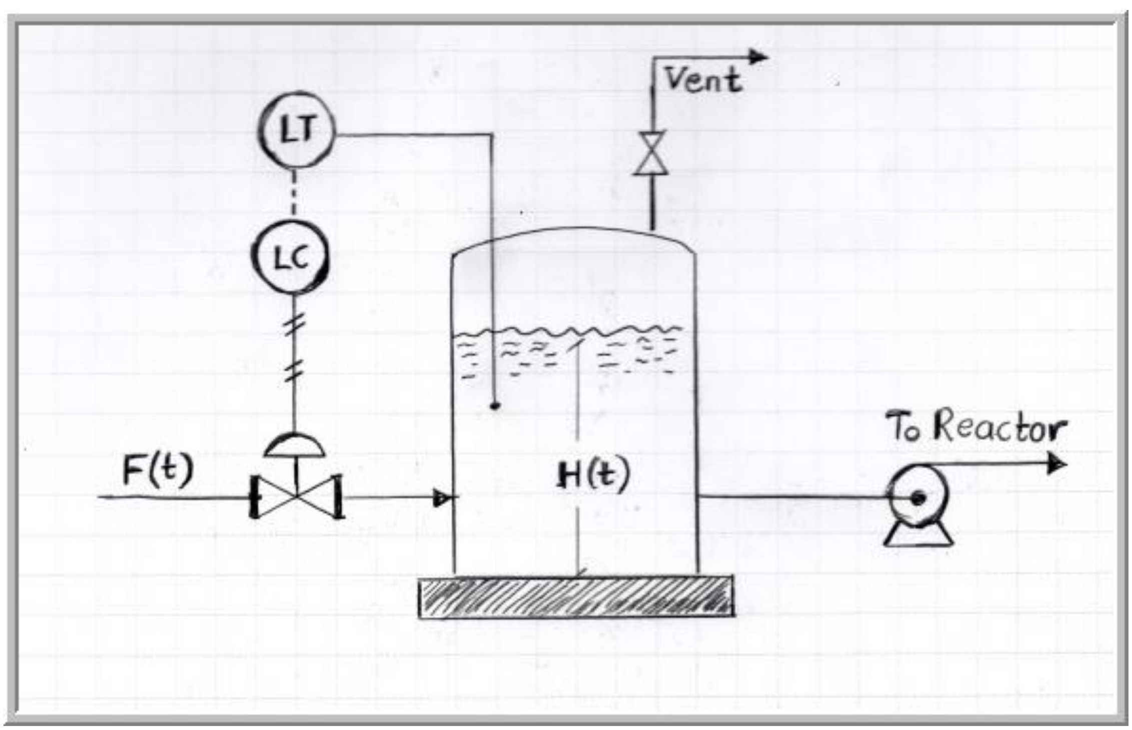
Problem 1: Design of PID Feedback Controller for Tank Level Control

The process model for a liquid storage system shown in the process diagram below is given by the following transfer function:

$$G_P(s) = \frac{H(s)}{F(s)} = \frac{K_P e^{-\theta s}}{s} \quad K_P = 0.2 \quad \theta = 7.4\,\text{min}$$

The objective is to control the level by manipulating the inlet flow rate. The outlet flow is directed to a chemical reactor and is kept constant under flow control using a constant pressure variable speed pump. The inlet flow rate acts both as the manipulated variable(MV) and as a disturbance(DV). A **PID controller** is considered for this task

A. Using the **I**nternal **M**ode **C**ontrol (IMC) tuning method derive expressions for the controller gain, reset and rate parameters
B. Draw a block diagram for this control loop and derive the transfer functions for the level response to (a) a set point change and (b) to a disturbance change
C. Using the **PIDController.xlsm** control loop simulator, prepare plots for the level response to unit step change in the disturbance variable for various values of the design parameter $\tau_C = \{4, 8, 16\}$. Based on your simulations select what you consider to be the most suitable design parameter. Calculate the settling times for all controllers.
D. Calculate the maximum process gain for which the PID controller you have designed is stable. Calculate an approximate value with the Routh criterion and the exact value using the frequency response analysis method.



Problem 1 Solution:

A. Using the Internal Mode Control (IMC) tuning method, derive expressions for controller gain, reset and rate parameters

Using the Padé approximation

$$e^{-s\theta} \approx \frac{1-\frac{s\theta}{2}}{1+\frac{s\theta}{2}} \qquad (1)$$

Transfer function from problem statement rewritten using above

$$G_p(s) \approx \frac{K_p\left(1-\frac{s\theta}{2}\right)}{s\left(1+\frac{s\theta}{2}\right)} \qquad (2)$$

Define $\tilde{G} = \tilde{G}_+ \tilde{G}_-$ where

$$\tilde{G}_+ = 1-\frac{s\theta}{2} \qquad (3)$$

$$\tilde{G}_- = \frac{K_P}{\left(1+\frac{s\theta}{2}\right)s} \qquad (4)$$

Construct controller G_C^* with low-pass filter $f = (\tau_C s+1)^{-r}$ ($r = 1$ for first- and second-order systems), such that

$$G_C^* = \frac{f}{\tilde{G}_-} \qquad (5)$$

For integrating processes (Such as the one provided in the problem statement)

$$f = \frac{(2\tau_C - C)s+1}{(\tau_C s+1)^2} \qquad (6)$$

$$C = \left.\frac{d\tilde{G}_+}{dS}\right|_{s=0} = -\frac{\theta}{2} \qquad (7)$$

Using the definitions of f and $\tilde{G}_-$ (4) in (5),

$$G_C^* = \frac{s\left(1+\frac{s\theta}{2}\right)\left(2\tau_C+\frac{\theta}{2}\right)s+1}{K_P\,(\tau_C s+1)^2} \qquad (8)$$

Controller transfer function is

$$G_c = \frac{G_c^*}{1 - G_c^* G_p} \tag{9}$$

Replacing G_c^* (8) and G_p (2) in (9) with some simplifications performed

$$Gc = \frac{2}{kp\left(\tau c + \frac{\theta}{2}\right)}\left[1 + \frac{1}{2\left(\tau c + \frac{\theta}{2}\right)s} + \frac{\left(\tau c\theta + \frac{\theta}{4}\right)^2}{2\left(\tau c + \frac{\theta}{2}\right)}\right] \tag{10}$$

From above,

Gain:

$$K_C = \frac{2}{kp\left(\tau c + \frac{\theta}{2}\right)}$$

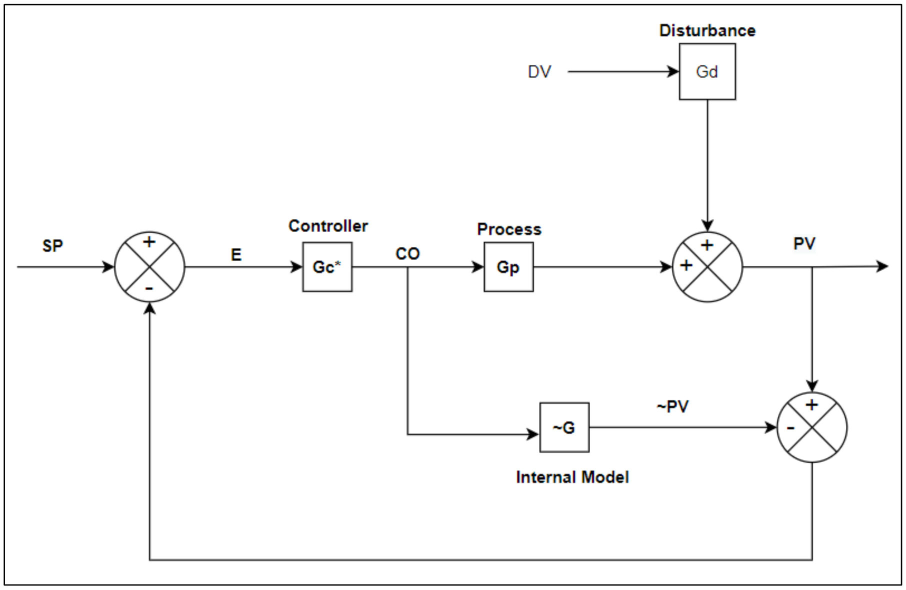
Reset:

$$\tau_I = 2\left(\tau_C + \frac{\theta}{2}\right)$$

Rate:

$$T_D = \frac{\left(\tau c\theta + \frac{\theta^2}{4}\right)}{2\left(\tau c + \frac{\theta}{2}\right)}$$

B. Draw a block diagram for this control loop and derive the transfer functions for the level response to (a) a set point change and (b) to disturbance change



SP = Set Point, E = Error, MV = Manipulated Variable, DS = Disturbance Variable, PV = Process Variable, G_C^* = Controller Transfer Function, G_P = Process Transfer Function, G_d = Disturbance Transfer function

Closed loop transfer function for level response with changes in setpoint and disturbances

$$PV = (MV)G_P + (DV)G_d \tag{11}$$

where

$$MV = EG_C = (SP - PV)G_C \tag{12}$$

Using (12) in (11) with rearrangement

$$PV = \frac{G_P G_C}{1 + G_P G_C}(SP) + \frac{G_d}{1 + G_P G_C}(DV) \tag{13}$$

The closed loop transfer function is then defined for (a) and (b) as

(a) Set point change (@ DV=0)

$$\boxed{\left.\frac{PV}{SP}\right|_{DV=0} = \frac{G_c G_p}{1 + G_c G_p}}$$

(b) Disturbance change (@ SP=0)

$$\boxed{\left.\frac{PV}{DV}\right|_{SP=0} = \frac{G_d}{1 + G_c G_p}}$$

C. Using the PIDController.xlsm control loop simulator, prepare plots for the level response to unit step change in the disturbance variable for various values of the design parameter $\tau_C = \{4, 8, 16\}$. Based on your simulations select what you consider to be the most suitable design parameter. Calculate the settling times for all controllers.

Using controller parameters found in **A.**, given process gain, time delay, and controller time delay, τ_C

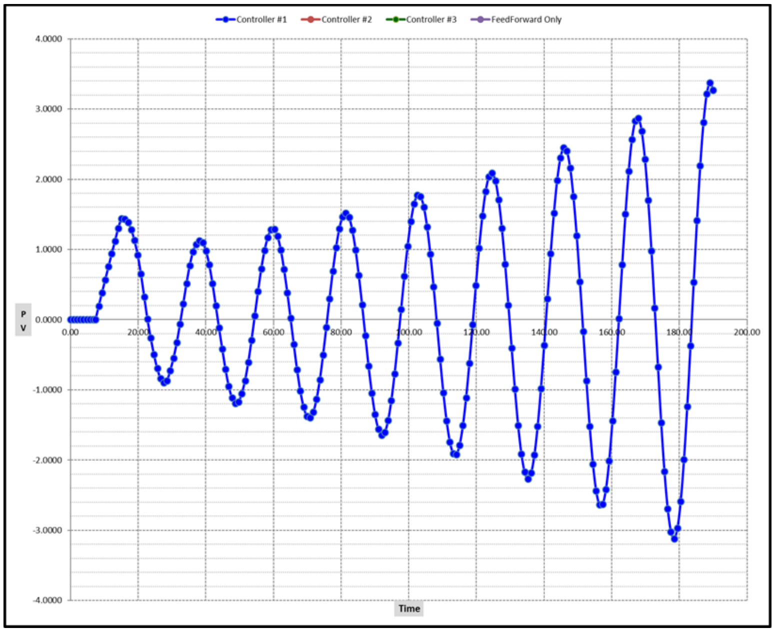
Controller	$\boldsymbol{\tau_C}$	$\boldsymbol{K_C}$	$\boldsymbol{\tau_I}$	$\boldsymbol{\tau_D}$
#1	4	1.299	15.4	2.811
#2	8	0.855	23.4	3.115
#3	16	0.508	39.4	3.353

Given process gain $K_P = 0.2$ and time delay $\theta_P = 7.4$ min, the value of the process gain K_P is updated to K_P^* using a range of lag times τ_p (100, 500, and 1000 minutes)

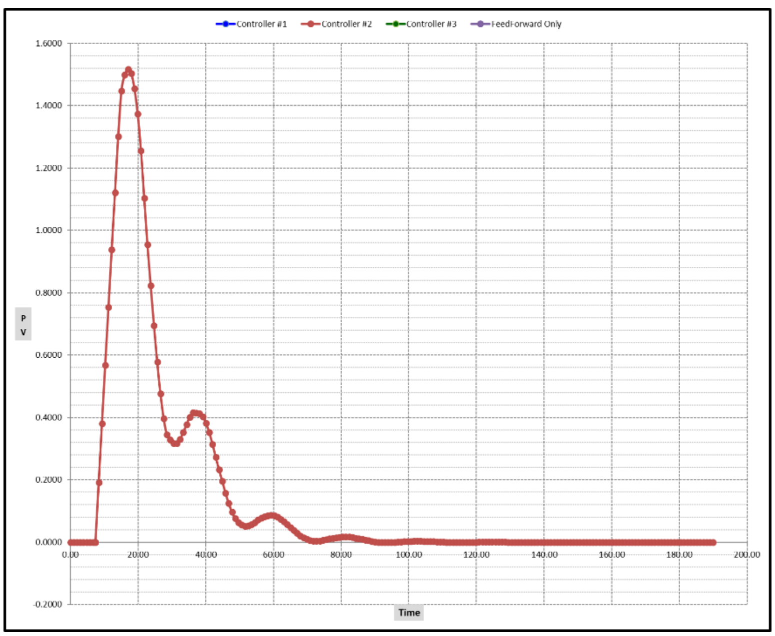
$$K_p^* = K_p \, \tau_p = 0.2 * 100 = 20$$

Using **PIDController.xlsm** to generate plots for level response to unit step change (Plots on following pages). Closed loop responses are more sluggish and less oscillatory for $\tau_C = 16$ compared with $\tau_C = 8$. Overshoot for $\tau_C = 16$ is less compared to others. For $\tau_C = 4$, oscillatory curve was obtained. **Controller #2** gives very short settling time with $\boldsymbol{\tau_C = 8}$. ***Controller 2 is the most suitable controller***

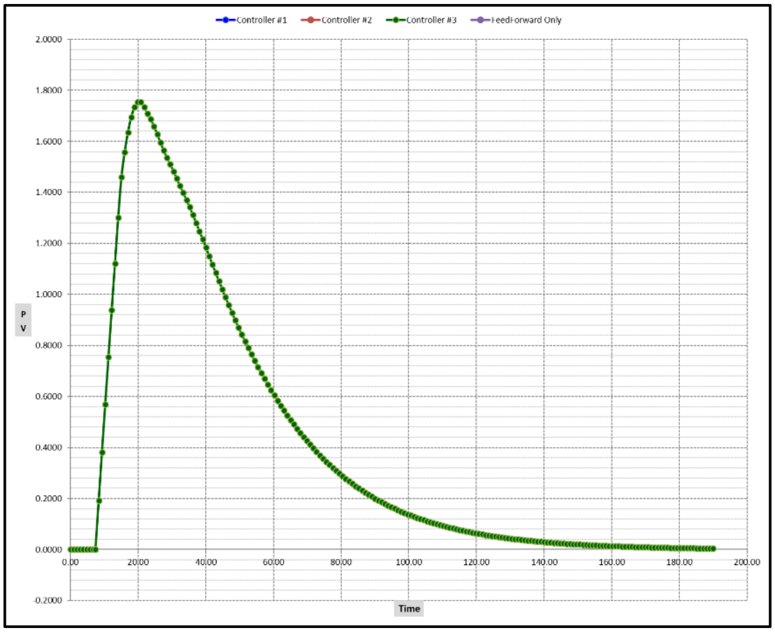
PV-Time for Controller #1, τ_C = 4. Controller oscillates with higher amplitudes each cycle therefore no settling time was achieved



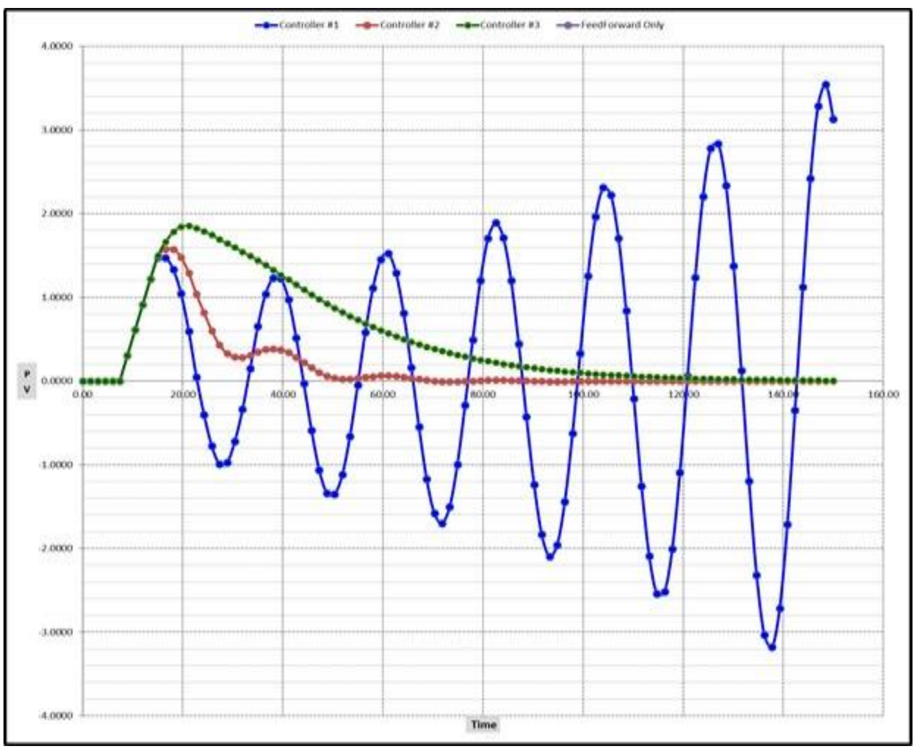
PV-Time for the Controller #2, τ_C = 8. Controller overshoots, oscillates and finally achieves steady state. Settling time calculated to be about 70 minutes



PV-Time for Controller #3, τ_C = 16. Controller shoots up, does not show an oscillatory behavior, then achieves steady state eventually. Settling time calculated to be about <u>133 minutes.</u>



PV-Time curves for all controllers overlayed.



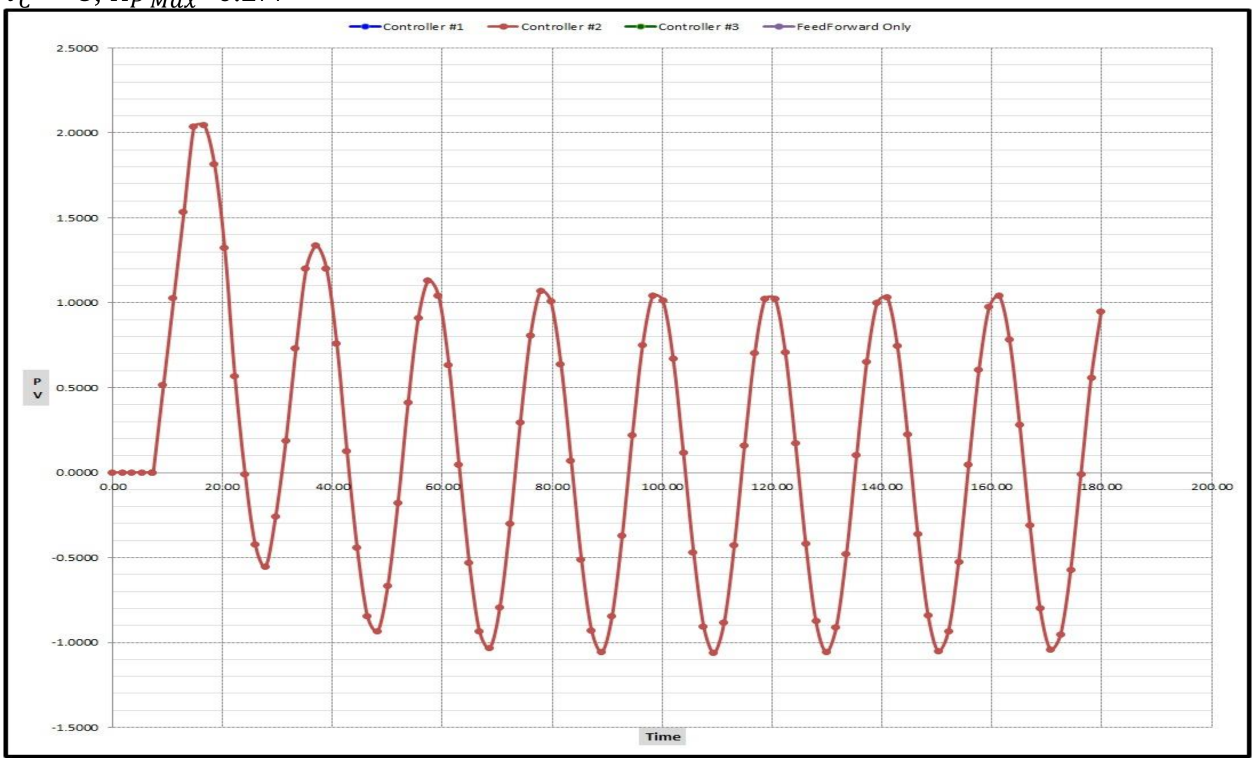
D. Calculate the maximum process gain for which the PID controller you have designed is stable. Calculate an approximate value with the Routh criterion and the exact value using the frequency response analysis method.

Controller #2 with $\tau_C = 8$ was found to be the most stable; maximum process gain calculated using its controller parameters.

Summarized parameters for controller #2

Controller	$\boldsymbol{\tau_C}$	$\boldsymbol{K_C}$	$\boldsymbol{\tau_I}$	$\boldsymbol{\tau_D}$
#2	8	0.855	23.4	3.115

Maximum process gain found to be 0.277. Plot below is the response of controller #2, $\tau_C = 8$, $K_{P\,Max}$=0.277



Routh criterion to approximate process gain

Characteristic equation for controller stability,

$$1 + G_C G_P = 0 \qquad (14)$$

G_C from before and G_P (2) in (14) gives

$$\left(\tau_I \frac{\theta}{2} - K_C K_P\, \tau_I\, \tau_D \frac{\theta}{2}\right) s^3 + \left(\tau_I + K_C K_P\, \tau_I \tau_D - K_C K_P \tau_I \frac{\theta}{2}\right) s^2 + \left(K_C K_P\, \tau_I - K_C K_P \frac{\theta}{2}\right) s + K_C K_P = 0 \qquad (15)$$

Making the following holders

$$a_0 = K_C K_P;\;\; a_1 = \left(K_C K_P\, \tau_I - K_C K_P \frac{\theta}{2}\right);\; a_2 = \tau_I + K_C K_P\, \tau_I \tau_D - K_C K_P \tau_I \frac{\theta}{2};$$

$$a_3 = \tau_I \frac{\theta}{2} - K_C K_P\, \tau_I\, \tau_D \frac{\theta}{2}$$

The array for (15) is

$$\begin{bmatrix} s^n & a_2 & a_0 \\ s^{n-1} & a_3 & a_1 \end{bmatrix}$$

$$b_1 = \frac{a_3 a_0 - a_2 a_1}{a_3},\quad b_2 = 0$$

$$c_1 = \frac{b_1 a_1 - b_2 a_3}{b_1}, c_2 = 0$$

Solving for b_1 and c_1 and simplifying,

$$b_1 = 3.007\, K_P - 6.81\, K_P^2$$

$$c_1 = 16.8365\, K_P$$

For stability, $b_1 > 0$,

$$3.007\, K_P - 6.81\, K_P^2 > 0$$

$$K_P < 0.4416$$

Similarly, $c_1 > 0$

$$16.8365\, K_P > 0$$

$$\mathrm{K_P} > 0$$

Approximated value for the process gain was found to be $\boxed{0 < K_P < 0.4416}$

Frequency response analysis method for the exact value of process gain

The expressions for G_P and G_C in (14),

$$G_p(s) = \frac{K_p e^{-s\theta}}{s}$$

$$G_c = K_C\left[1 + \frac{1}{\tau_I s} + \tau_D s\right]$$

Replacing these into the equation with some simplifications produces

$$\tau_I s^2 + K_P K_C e^{-s\theta}(1 + \tau_I s + \tau_D \tau_I s^2) = 0 \qquad (16)$$

Applying the Bode Stability Criterion and substituting $s = i\omega$, where $i = (-1)^{1/2}$,

$$-\tau_I \omega^2 + K_P K_C e^{-i\omega\theta}[1 + \tau_I (i\omega) - \tau_D \tau_I \omega^2] = 0 \qquad (17)$$

Applying Euler's formula, $e^{-i\omega\theta} = cos(\omega\theta) - isin(\omega\theta)$, (17) becomes

$$-\tau_I \omega^2 + K_P K_C[cos(\omega\theta) - isin(\omega\theta) + i\tau_I \omega\, cos(\omega\theta) + \tau_I \omega sin(\omega\theta) - \tau_D \tau_I \omega^2\, cos(\omega\theta) + i\tau_D \tau_I \omega^2 sin(\omega\theta)] = 0$$

Equating real parts of the equation to zero,

$$-\tau_I \omega^2 + K_P K_C[cos(\omega\theta) + \tau_I \omega sin(\omega\theta) - \tau_D \tau_I \omega^2\, cos(\omega\theta)] = 0 \qquad (18)$$

Similarly, for the imaginary part,

$$K_P\ K_C\ i\ [sin(\omega\theta) + \tau_I\omega\ cos(\omega\theta) + \tau_D\tau_I\omega^2 sin(\omega\theta)] = 0 \qquad (19)$$

Rewriting above,

$$[sin(\omega\theta) + \tau_I\omega\ cos(\omega\theta) + \tau_D\tau_I\omega^2 sin(\omega\theta)] = 0 \qquad (20)$$

Solving for ω,

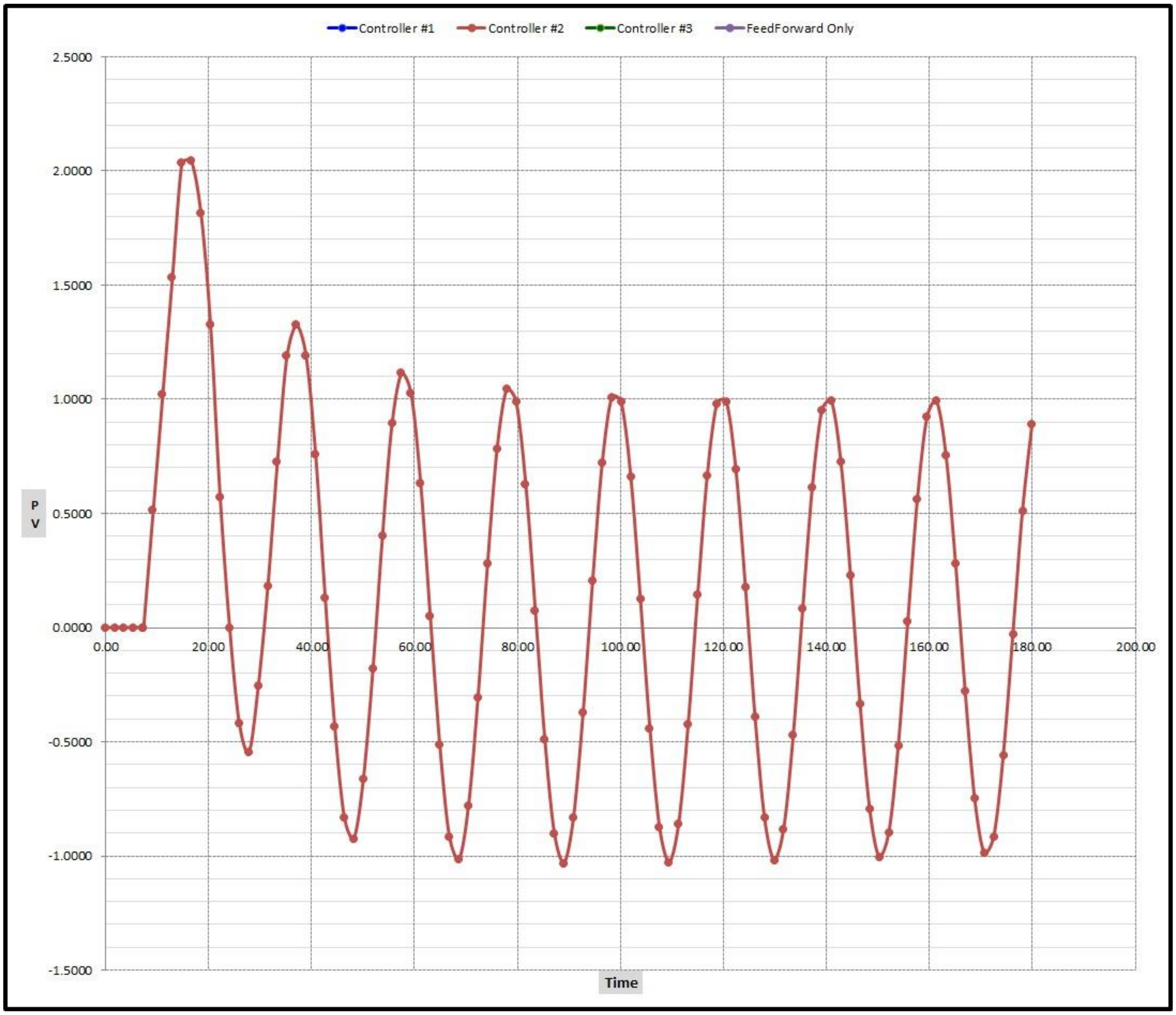
$$\boxed{\omega = 0.304}$$

Plugging ω into (19) gives

$$\boxed{K_{P\ Max} = 0.277}$$

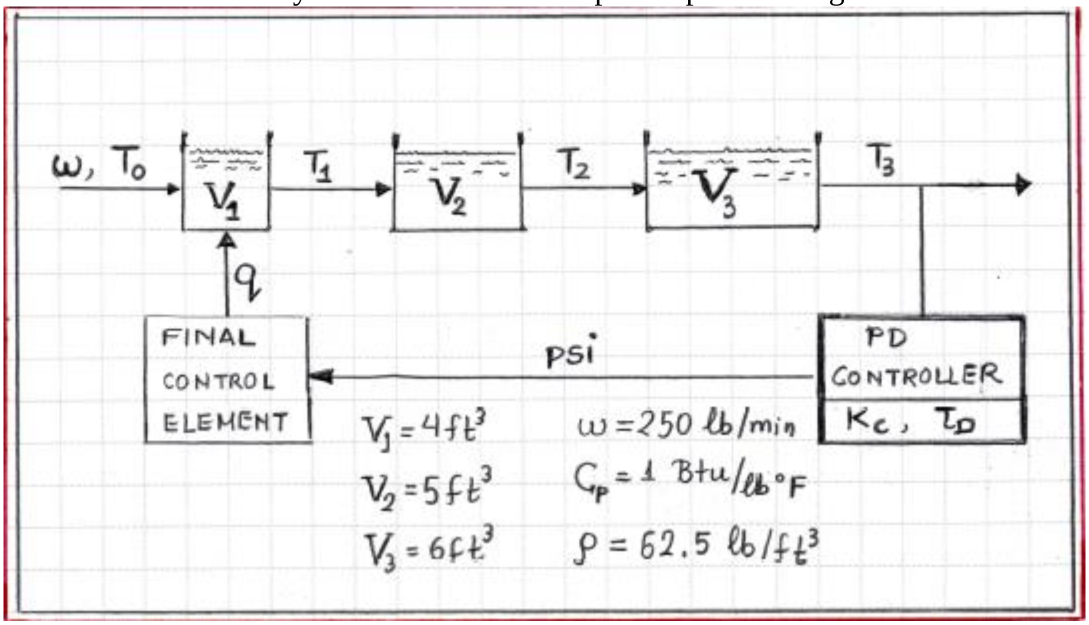
Value of $K_{P\ Max}$ implemented into Excel calculator to visualize Controller #2 response. Plot below demonstrates sustained oscillations.

PV-T Controller response for $\tau_C = 8$, $K_{P\ Max}$=0.277.


Controller #1
Controller #2
Controller #3
FeedForward Only
2.5000
2.0000
1.5000
1.0000
0.5000
0.0000
-0.5000
-1.0000
-1.5000
PV
0.00
20.00
40.00
60.00
80.00
100.00
120.00
140.00
160.00
180.00
200.00
Time

Problem 2: High Order Systems and Skögestad Approximations

Consider the thermal system shown in the simplified process diagram below:



The thermal system is currently being controlled by a **P**roportional **D**erivative (PD) controller with gain and rate equal to: $\boxed{K_C = 250\dfrac{\text{Btu/min}}{^{o}\text{F}} \qquad \tau_D = 0.5\,\text{min}}$ The temperature T_o of the inlet stream is considered a disturbance.

A. Draw a block diagram of the control system with the transfer functions in each block. Each transfer function must only have the numerical values of its parameters.
B. From this block diagram, determine the closed loop transfer function relating the temperature in tank #3 to a change in the set point (heat flow to the first tank).
C. From this block diagram determine the closed loop transfer function relating the temperature in tank #3 to a change in the inlet temperature T_o
D. Calculate the offset for a unit step change in the inlet temperature T_0 by applying the final value theorem
E. Approximate the higher order transfer functions with FOPTDs and perform a simulation using the **PIDController.xlsm** Excel tool and see whether you can verify your theoretical answer in question #4.
F. You have been asked to eliminate the observed offset. Design a controller and calculate its parameters that would accomplish just that. Perform a simulation again, using the **PIDContoller.xlsm** Excel tool to verify that the observed offset has been eliminated and calculate the settling time.

PROBLEM 2. Skogestad Approximations & Final value theorem

A. We begin by developing the process & disturbance transfer functions

Tank #1 Disturbance variable is the feed temperature T_0
Manipulate variable is the heat flow q
Process Variable is the effluent temperature

We now perform an energy balance around tank #1

$$\rho V_1 C_p \frac{dT_1}{dt} = w C_p (T_0 - T_1) + q \quad \text{Define } \frac{\rho V_1}{w} = \tau_1 \quad \frac{q}{wC_p} = \frac{q}{250} \Rightarrow$$

$$\tau_1 \frac{dT_1}{dt} = T_0 - T_1 + \frac{q}{250} \quad (\text{Eq. } 1)$$

At steady-state $\tau_1 \frac{dT_1^{ss}}{dt} = T_0^{ss} - T_1^{ss} + \frac{q^{ss}}{250}$ (Eq. 2)

Define the deviation variables:

Process Variable Deviation: $Y_1 = T_1 - T_1^{ss}$ Process Gain $K_p = \frac{1}{250}$

Manipulated Variable Deviation: $X = q - q^{ss}$

Disturbance Variable Deviation $D = T_0 - T_0^{ss}$.

Subtracting (Eq. 2) from (Eq. 1) and using deviation variables

$$\tau_1 \frac{dY_1}{dt} + Y_1 = K_p X + D \quad \text{since } \tau_1 = \frac{\rho V_1}{w} = \frac{(62.5)(4)}{250} = 1 \text{ min}$$

$$\frac{dY_1}{dt} + Y_1 = K_p X + D \Rightarrow s Y_1(s) + Y_1(s) = K_p X(s) + D(s) \Rightarrow \boxed{Y_1(s) = \frac{K_p}{s+1} X(s) + \frac{1}{s+1} D(s)}$$

Therefore from the analysis of tank #1 we obtain its process transfer function G_{P1} and disturbance transfer function G_d where

$$G_{P1} = \frac{Y_1(s)}{X(s)} = \frac{0.004}{s+1} \qquad G_d = \frac{Y_1(s)}{D(s)} = \frac{1}{s+1}$$

Tanks #2 and #3

Writing energy balances for Tanks #2 and #3 we get:

$$\rho V_2 \, C_p \frac{dT_2}{dt} = w C_p (T_1 - T_2) \quad \text{and} \quad \rho V_3 \frac{dT_3}{dt} = w C_p (T_2 - T_3)$$

Introducing again deviation variables $Y_2 = T_2 - T_2^{ss}$, $Y_3 = T_3 - T_3^{ss}$ we obtain

$$\frac{\rho V_2}{w} \frac{dY_2}{dt} = Y_1 - Y_2 \Rightarrow \tau_2 \frac{dY_2}{dt} + Y_2 = Y_1 \Rightarrow \frac{Y_2(s)}{Y_1(s)} = \frac{1}{\tau_2 s + 1}$$

Since $\tau_2 = \frac{\rho V_2}{w} = \frac{(62.5)(5)}{250} \Rightarrow \tau_2 = 1.25 \text{ min}$

and therefore $G_{P2} = \frac{Y_2(s)}{Y_1(s)} = \frac{1}{1.25s+1}$

Following similar steps for tank #3 we obtain

$$\frac{Y_3(s)}{Y_2(s)} = \frac{1}{\tau_3 s + 1} \quad \text{and since} \quad \tau_3 = \frac{\rho V_3}{w} = \frac{(62.5)\,6}{250} \Rightarrow \tau_3 = 1.5$$

we get $G_{P3} = \frac{Y_3(s)}{Y_2(s)} = \frac{1}{1.5s+1}$

To summarize the overall process transfer function is $G_p = G_{p1}\, G_{p2}\, G_{p3} \Rightarrow$

$$G_p = \frac{Y_3(s)}{Q(s)} = \frac{K_p(=0.004)}{(1.5s+1)(1.25s+1)(s+1)}$$

Setting $Y_3(s) = PV$ (Process (controlled) Variable)
$Q(s) = MV$ (Manipulated Variable)
$T_0(s) = DV$ (Disturbance Variable)

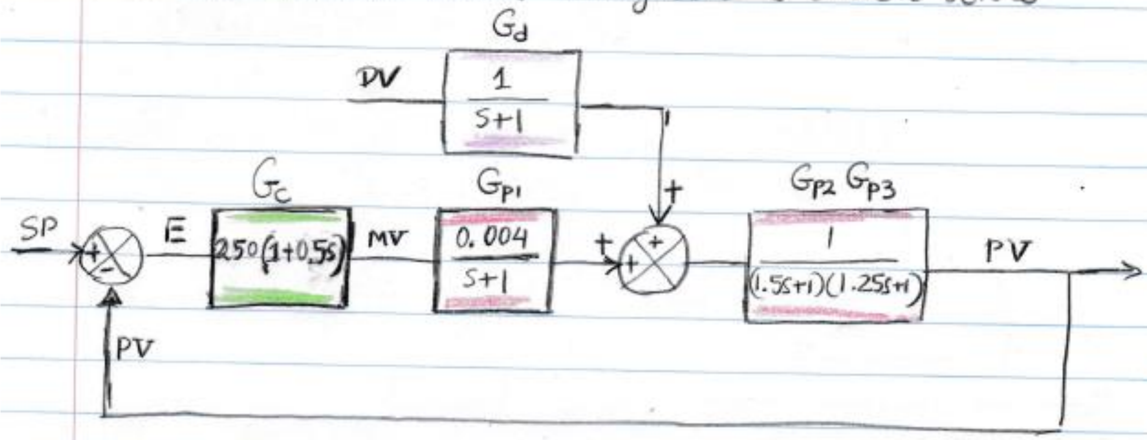
we get

$$G_p(s) = \frac{PV(s)}{MV(s)} = \frac{K_p(=0.004)}{(1.5s+1)(1.25s+1)(s+1)} \qquad G_d(s) = \frac{PV}{DV} = \frac{1}{s+1}$$

The Controller Transfer function is

$$G_c = K_c(1+\tau_D s) \Rightarrow G_c = 250(1+0.5s)$$

and the Control Block Diagram is drawn below



B. Closed Loop Transfer function $\frac{PV}{SP}$ (Set DV=0)

Looking at the control block diagram we have

$$PV = (MV) \cdot G_{p1} G_{p2} G_{p3} = (MV) \cdot G_p$$

$$MV = (SP - PV)\, G_C$$

Combining the two equations

$$PV = (SP - PV)\, G_C G_p \Rightarrow \frac{PV}{MV} = \frac{G_C G_p}{1 + G_C G_p} \Rightarrow$$

$$\frac{PV}{SP} = \frac{250(1+0.5s)\frac{0.004}{(1.5s+1)(1.25s+1)(s+1)}}{1 + 250(1+0.5s)\frac{0.004}{(1.5s+1)(1.25s+1)(s+1)}} \Rightarrow$$

$$\boxed{\frac{PV}{SP} = \frac{0.5s+1}{(1.5s+1)(1.25s+1)(s+1) + (0.5s+1)}} \quad (\text{Eq. } 1)$$

C. Closed Loop Transfer function $\frac{PV}{DV}$ (Set SP=0)

$$PV = [(MV)\, G_{p1} + (DV) G_d]\, G_{p2} G_{p3}$$

$$MV = -(PV)\, G_C$$

Combining the two equations we obtain

$$PV = [-(PV)\, G_C G_{p1} G_{p2} G_{p3}] + (DV)\, G_d G_{p2} G_{p3} \Rightarrow$$

$$PV[1 + G_C G_{p1} G_{p2} G_{p3}] = DV \; G_d G_{p2} G_{p3} \Rightarrow$$

$$\frac{PV}{DV} = \frac{G_d G_{p2} G_{p3}}{1 + G_C G_p} \Rightarrow \frac{PV}{DV} = \frac{\frac{1}{(1.5s+1)(1.25s+1)(s+1)}}{\frac{(1.5s+1)(1.25s+1)(s+1) + (0.5s+1)}{(1.5s+1)(1.25s+1)(s+1)}} \Rightarrow$$

$$\frac{PV}{DV} = \frac{1}{(1.5s+1)(1.25s+1)(s+1) + (0.25s+1)}$$

D. From the closed loop transfer function we get for a unit step in the disturbance

$$PV(s) = \frac{1}{S}\,\frac{1}{(1.5s+1)(1.25s+1)(s+1)+(0.5s+1)}$$

Applying the final value theorem we have

$$PV(t\to\infty) = \lim_{s\to 0} s\cdot PV(s) \Rightarrow \boxed{PV(t\to\infty) = 0.5}$$

E. To use the CHE341 Calculator we have the following Data

Disturbance: Gain $=1$

Lag Time $=1$

Dead Time $=0$

Disturbance active on CONTROL VALVE OUTPUT

Control Valve: Gain $= 0.004$

Lag Tim $= 1$

Controller: Gain $= 250$

Integral Time $= 0$

Derivative Time $= 0.5$

PROCESS:

Transfer function $G_{P2}\,G_{P3} = \dfrac{1}{(1.5s+1)(1.25s+1)}$

Applying the Skogestad approximation

$$\tau^* = 1.5 + \frac{1.25}{2} = 2.125 \text{ and } \theta^* = 0.625 \Rightarrow G_{P2}G_{P3} \approx \frac{e^{-0.625}}{2.125s+1}$$

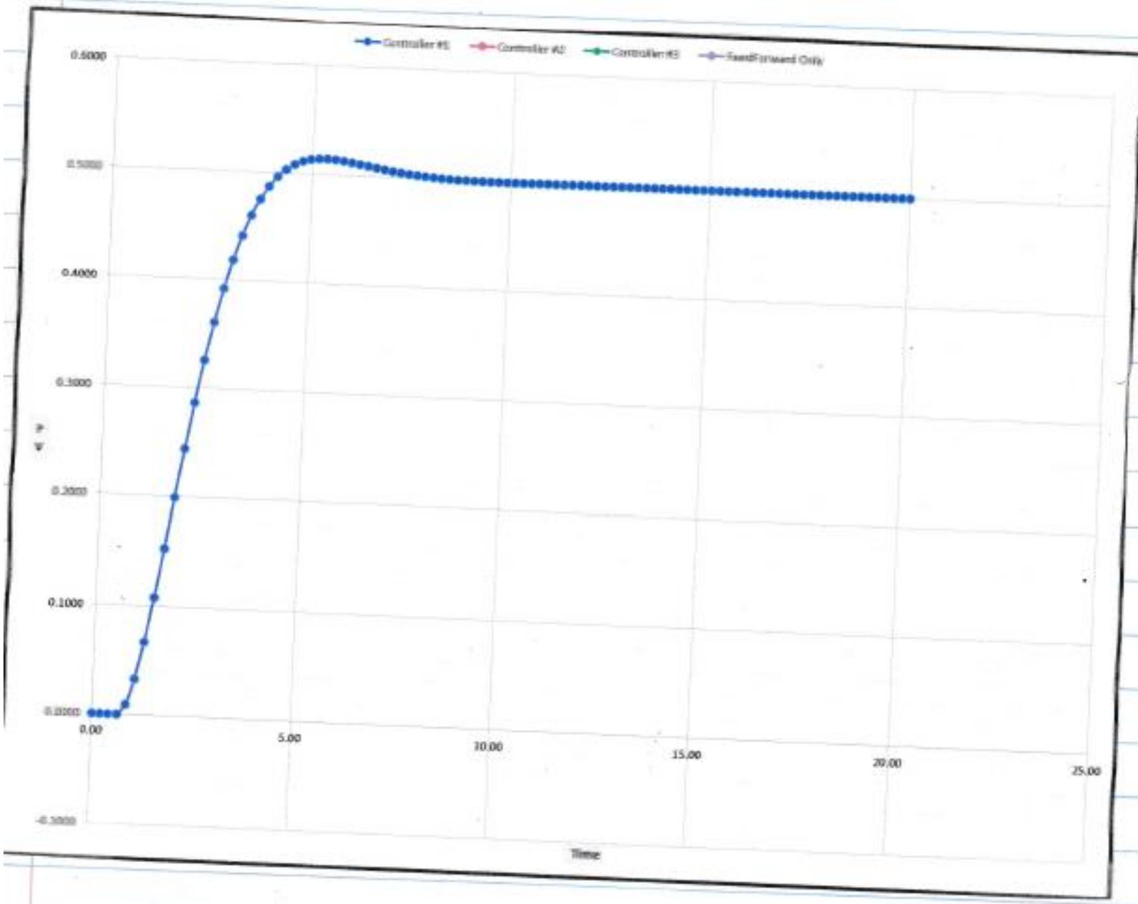
Therefore for the process in the ChE Calculator we have

Process: Gain = 1

Lag Time = 2.125

Dead time = 0.625

When we enter all data in the ChE341 Calculator we obtain the graph shown below that verifies our answer in Part D.



F. To design a controller that eliminates the offset we first must consider the overall process transfer function $G_p = \dfrac{0.004}{(1.5s+1)(1.25s+1)(s+1)}$

Now we use Skögestadt's rule to develop an FOPT model

$$\tau^* = 1.5 + 1.25/2 = 2.125$$

$$\Theta^* = 1.25/2 + 1 = 1.625$$

We now use these values to design a PI controller using the IMC tuning method with $\tau_c = \Theta^*$ we obtain $K_c = 163.4615$

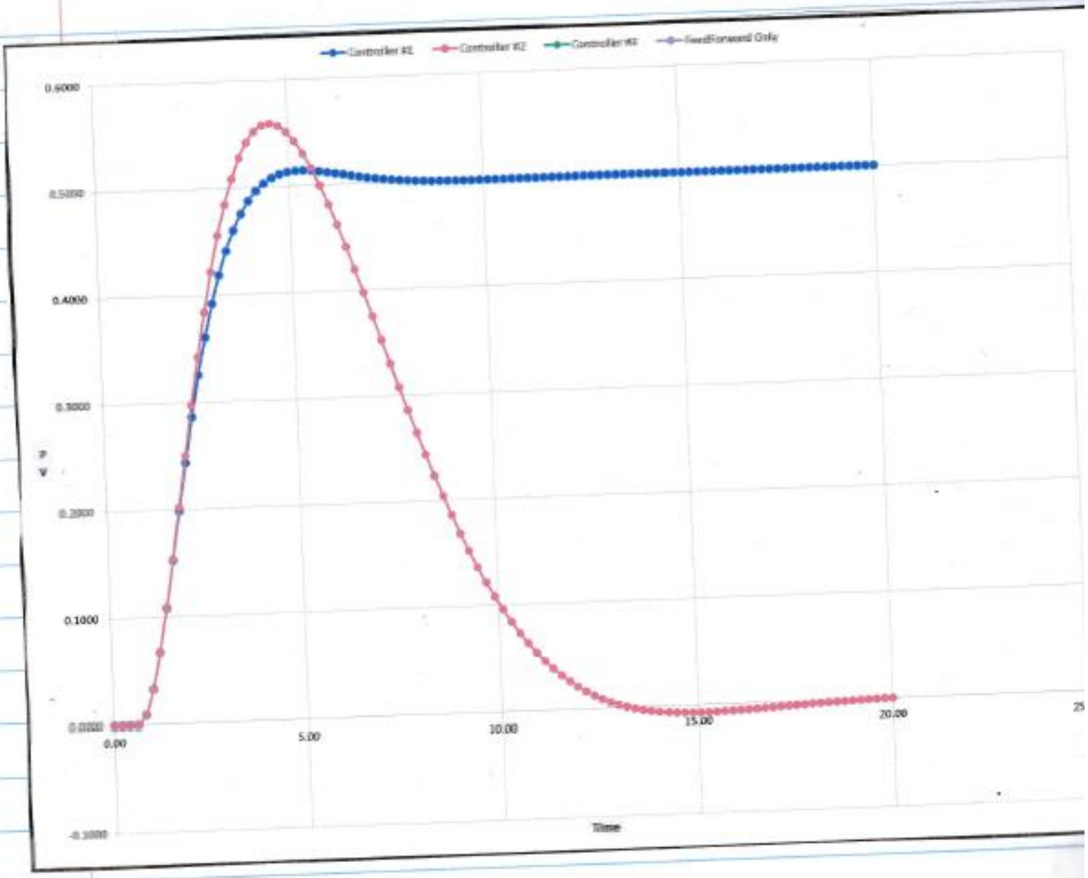
$$\tau_I = 2.125$$

Running the simulation with exactly the same input as in part D but now with the new controller we obtain the graph shown below

We observe that the PI rejects the disturbance nicely with no offset and a small overshoot

The settling time is around 12 min. For faster settling time we may consider adding some derivative action. If the overshoot is a problem we may consider implementing a Feed forward controller since the effect of the disturbance on the tank temperature is well understood.

Plots of PD Controller (Blue Line) with offset and PI Controller (Red line) with the offset eliminated.



Problem 3: **Ratio Controller for a Waste Treatment Facility**

A pH neutralization process is shown in the process flow diagram below. The effluent is fed to a pond where biological treatment takes place. Without control changes in the acid feed flow rate can cause significant excursions to the pH of the solution that can seriously affect the biological treatment of the waste water by altering the ability of the biological agents to remove the organic matter. Ratio control is used to manage the pH in these types of situations. Ratio control is a type of feedforwad control with a gain, dead time and a lead-lag element.

A. For this neutralization problem define the process(controlled) variable, the wild(disturbance) and manipulated variables and draw a process control block diagram.

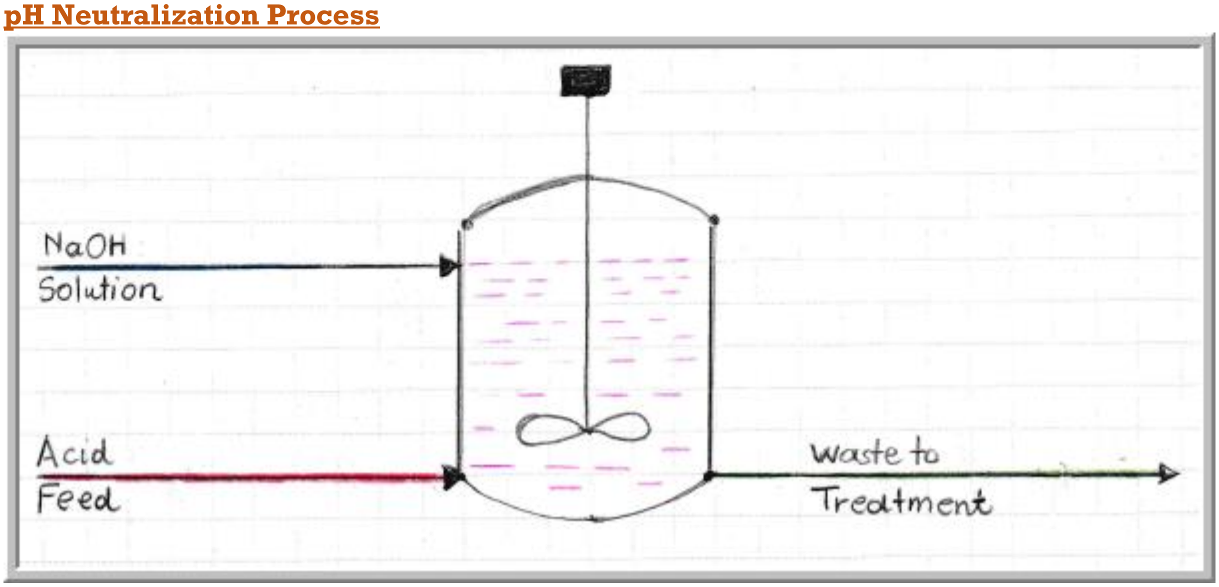
B. Develop the closed loop transfer function the relates the changes to the process variables to changes to the wild(disturbance variable) flow.

C. Complete the process control block diagram with a feedback controller that maintains the pH of the effluent at a specified set point. Develop the new closed loop transfer function that relates the changes of the controlled variable (pH) to the changes in the wild(disturbance variable) flow.

D. Complete the P&ID Diagram showing all the controllers and sensors

Note: The wild flow is also known as master flow

pH Neutralization Process



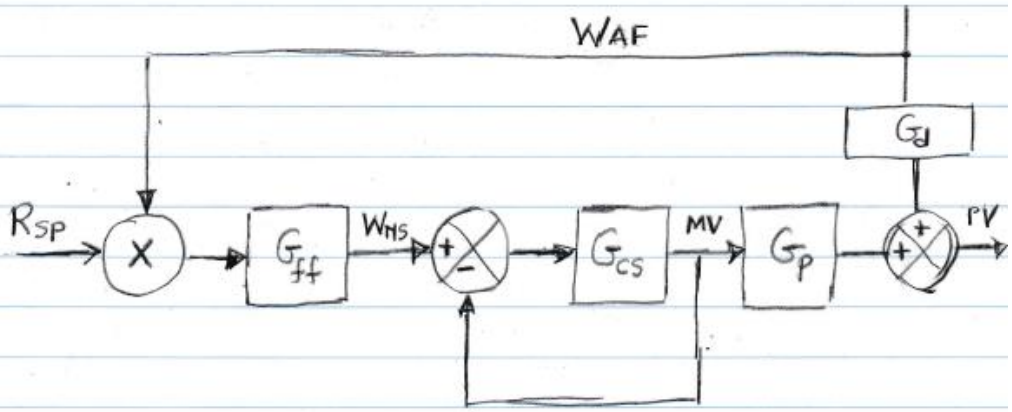
Problem 3. Ratio Controller for Wastewater Treatment

A. Process (Controlled) Variable: pH of effluent
Disturbance (Wild) Variable: Acid Feed W_{AF}
Manipulated (Controlled) Variable: NaOH Solution W_{NS}

The NaOH solution flow is considered also as controlled variable, because the signal from the Ratio Controller is cascaded to a flow control loop. Consider also the following Definitions of Transfer functions

1. Process (Blending) Transfer function: G_P
2. Disturbance Transfer function: G_d
3. Ratio (Feed Forward) Controller Transfer function: G_{FF}
4. Cascade Flow Controller Transfer function: G_{CS}
5. Feedback Controller Transfer function: G_C
6. Ratio Controller Set Point: RSP
7. Feedback Controller Set Point: SP

Based on the above definitions we can now draw the Control Block Diagram of the Ration Controller without Feedback



B. Development of Closed Loop Transfer Function

$PV = G_d(W_{AF}) + G_p(MV)$ (eq. 1)

$MV = (W_{NS} - MV)G_{CS} \Rightarrow MV = \dfrac{W_{NS}G_{CS}}{1 + G_{CS}}$ (eq. 2)

$W_{NS} = R_{SP} W_{AF} G_{FF}$ (Eq. 3)

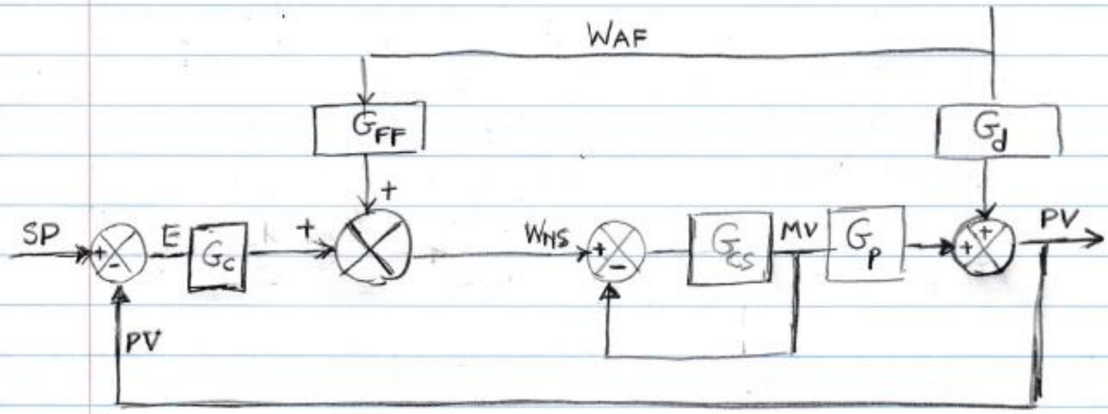
Combining (Eq. 2) & (Eq. 3) we obtain $MV = \dfrac{R_{SP} W_{AF} G_{FF} G_{CS}}{1 + G_{CS}}$ (Eq. 4)

Inserting (Eq. 4) into (Eq. 1) we obtain

$$PV = G_d(W_{AF}) + G_p \frac{R_{SP}(W_{AF})G_{FF}G_{CS}}{1 + G_{CS}}$$

$$\boxed{\frac{PV}{W_{AF}} = \frac{PV}{DV} = \frac{G_d(1 + G_{CS}) + R_{SP}G_{FF}G_pG_{CS}}{1 + G_{CS}}}$$ (Eq. 5)

C. We now introduce Feedback to the Ratio Control loop
The new Control Block Diagram is shown below



To derive the closed loop transfer function observe that everything remains the same as in Case B. but RSP now is set by the feedback controller, i.e.

$$W_{NS} = (SP - PV)\,G_C + G_{FF}\,W_{AF} \quad (\text{Eq. } 6)$$

Since we are only interested in the control loop transfer function that relates PV to DV($=W_{AF}$) we set $SP=0$ and substituting (Eq. 6) into (Eq. 2) we obtain

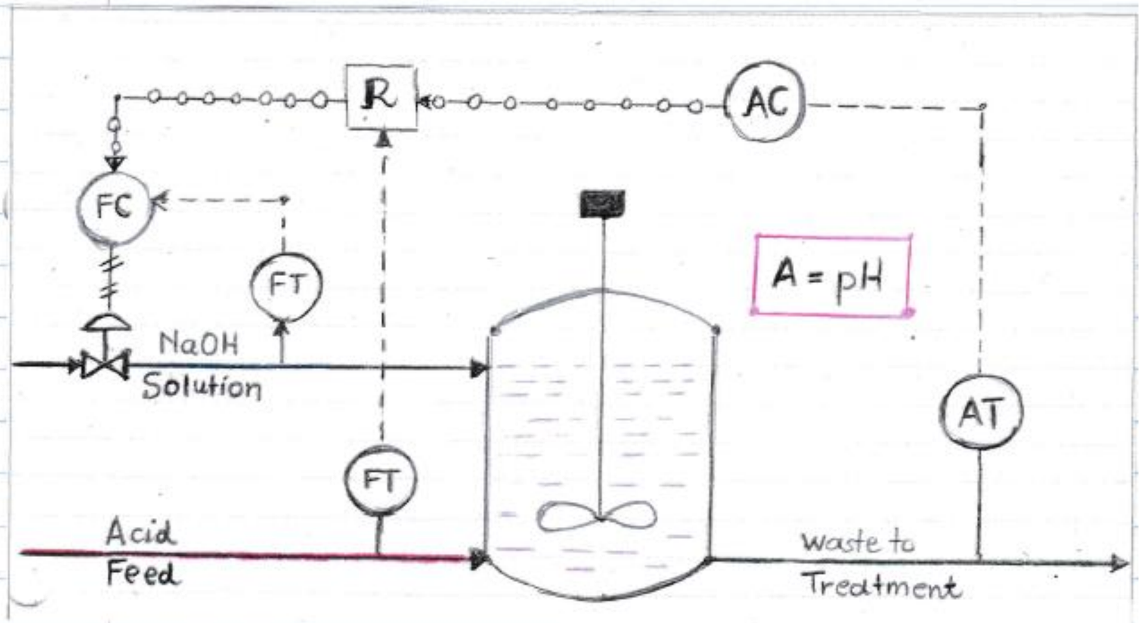
$$MV = \frac{(-PV\,G_C + G_{FF}\,W_{AF})\,G_{CS}}{1 + G_{CS}} \quad (\text{Eq. } 7)$$ and inserting (Eq. 7) into (Eq. 1) we obtain

$$PV = G_d\,W_{AF} + \frac{-PV\,G_C + G_{FF}W_{AF}}{1 + G_{CS}} \cdot G_{CS}\,G_P \Rightarrow$$

$$PV\left(1 + \frac{G_C\,G_{CS}\,G_P}{1 + G_{CS}}\right) = W_{AF}\left(G_d + \frac{G_{FF}\,G_{CS}\,G_P}{1 + G_{CS}}\right) \Rightarrow$$

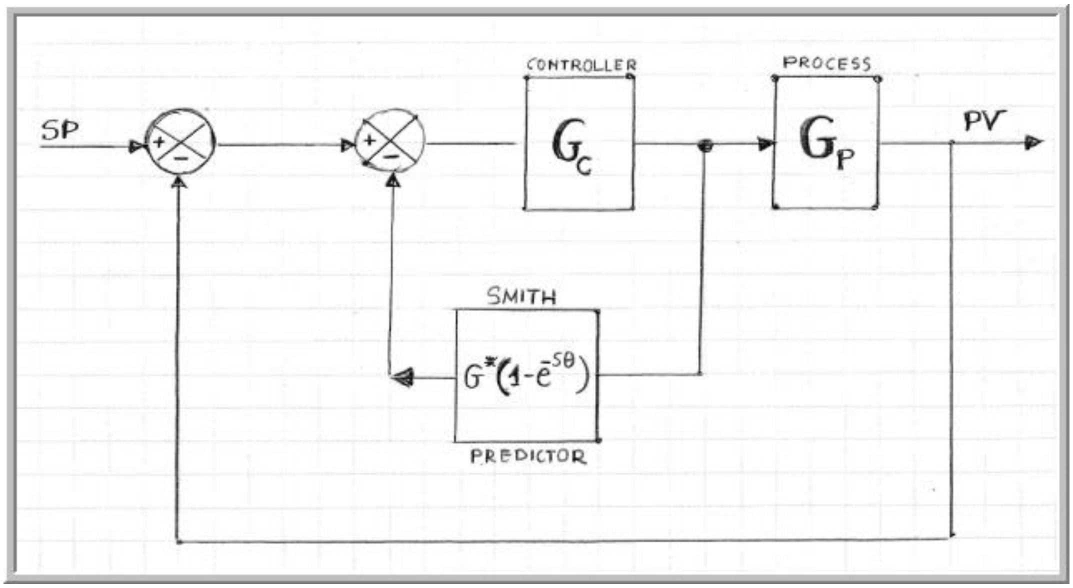
$$\frac{PV}{W_{AF}} = \frac{PV}{DV} = \frac{G_d(1+G_{CS}) + G_{FF}G_{CS}G_P}{1+G_{CS}(1+G_CG_P)}$$

D. A P&ID diagram of the process with the Feedback, FeedForward and Cascade Control for the pH neutralization process is given below



Problem 4: Time Delay Compensation with the Smith Predictor

Time delays frequently occur in the process industries because of the presence of transportation lags, recycle loops and sensor inertia. Time delays result in reduction of the controller gain in order to maintain stability and as a result make the controller performance sluggish. To improve the performance of loops with time delays, advanced control strategies have been developed over the years. On such strategy is known as the Smith predictor and it consists of an ordinary feedback loop plus an inner loop that introduces an extra term directly into the feedback path. A block diagram of the Smith predictor controller is given below:



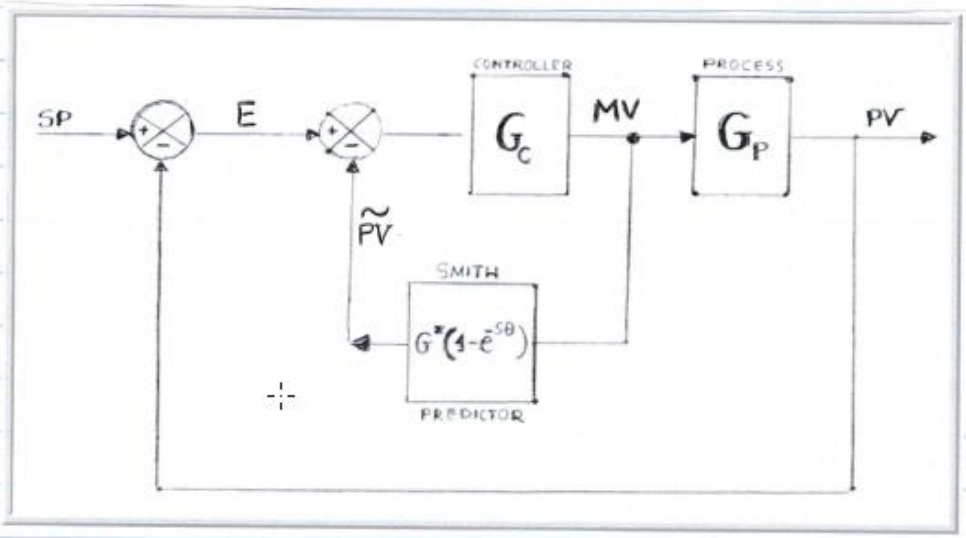
Where $G_P(s) = G^*(s)e^{-s\theta}$ and $G^*(s)$ is the process transfer function free of time delays.

A. Derive the closed-loop transfer function $\dfrac{PV(s)}{SP(s)}$

B. Consider the following second order process $G_P(s) = \dfrac{e^{-2s}}{(5s+1)(3s+1)}$. This chemical process is controlled with a PI feedback controller with $\{K_C = 1.23 \text{ and } \tau_I = 7.0\,\text{min}\}$. For the same second order process but with time delay equal to zero a PI feedback controller with constants $\{K_C = 3.02 \text{ and } \tau_I = 6.5\,\text{min}\}$ is used.

1. Show the response of the process <u>with</u> the time delay to a step change in the set point and calculate the settling time.
2. Show the response for the process <u>without</u> the time delay equal to zero and calculate the settling time again.
3. Derive and plot the Smith predictor response for the process <u>with</u> the time delay and calculate the settling time. Did the Smith predictor improve the controller performance? If so by how much?
4. Show all three responses on the same graph for better visual inspection and user interpretation

Problem 4. Time Delay Compensation with Smith-Predictor



A. We designate PV = Process Variable $\quad SP$ = Set Point

MV = Manipulated Variable $\quad E = SP - PV$

$\widetilde{PV}$ = Smith Predictor Signal

Using the properties of transfer functions we obtain

$(PV) = G_p\,(MV)$ (Eq. 1)

$(MV) = G_C\,(SP-PV) - G_c\,(\widetilde{PV})$ (Eq. 2)

$(\widetilde{PV}) = G^*(1-e^{-s\theta})(MV)$ (Eq. 3)

Substituting (Eq. 3) into (Eq. 2) and solving for (MV) we get

$$(MV) = \frac{G_c\,(SP-PV)}{1+G_c G^*(1-e^{-s\theta})} \qquad (\text{Eq. } 4)$$

Substituting (Eq. 4) into (Eq. 1) we obtain

$$PV = \frac{G_c\,G_p\,(SP-PV)}{1+G_c G^*\,(1-e^{-s\theta})} \Rightarrow PV\left[1+G_c G^*(1-e^{-s\theta})+G_c G_p\right] = G_c G_p\,(MV) \Rightarrow$$

$$\frac{PV}{SP} = \frac{G_c G_p}{1+G_c G^* - G_c G^* e^{-s\theta} + G_c G_p} \Rightarrow$$

$$\frac{PV}{SP} = \frac{G_c G^* e^{-s\theta}}{1+G_c G^* - G_c G_p + G_c G_p} \Rightarrow \boxed{\frac{PV}{SP} = \frac{G_c G^*}{1+G_c G^*} e^{-s\theta}} \quad \text{(Eq.5}$$

B. For the process $G_p = \dfrac{e^{-2s}}{(5s+1)(3s+1)}$ we have

$$\boxed{G^* = \frac{1}{(5s+1)(3s+1)}}$$

1. To generate the plot of the response of the actual process with time delay we set

Process Gain = 1 Valve Gain = 1 Controller Gain = 1.23
Lag Time = 5 Valve Lag Time = 3 Integral Time = 7.00
Time Delay = 2

The generated plot is shown with blue line: Controller 1

2. To generate the plot of the response of the process with time delay equal to zero we set

Process Gain = 1 Valve Gain = 1 Controller Gain = 3.02
Lag Time = 5 Valve Lag Time = 3 Integral Time = 6.5
Time Delay = 0

The generated plot is shown with red line: Controller 2

Settling Times: Actual Process ≈ 32 min
Process with Zero Time Delay ≈ 19 min

3. To Generate the response for the Smith predictor observe from equation (5) that

$$\frac{PV}{SP} = \frac{G_C G^*}{1+G_c G_c^*} e^{-s\theta}$$

For a unit step change $SP = \frac{1}{s}$ and therefore

$$PV = \frac{1}{s} \frac{G_c G^*}{1+G_c G^*} e^{-s\theta}$$

we designate $F(s) = \frac{1}{s} \frac{G_c G^*}{1+G_c G^*}$ then

$$PV = F(s) e^{-s\theta} \Rightarrow PV(t) = \mathcal{L}^{-1}\left[F(s) e^{-s\theta}\right] \Rightarrow$$

$$PV(t) = F(t-\theta)$$

However $F(t)$ has already been obtained so we simply shift it by 2 min and that is easily accomplished in Excel since we chose the time intervals to be exactly equal to $\Delta t = 1$ min

The generated plot is shown with the green line: Controller 3

4. All plots are shown in the graph below

Settling time with the Smith predictor is about 21 min and therefore the improvement is $\sim 34\%$ with no extra overshoot.

Generated plots for

1. Actual Process (Blue Line: Controller 1)
2. Process with zero delay (Red Line: Controller 2)
3. Actual Process with Smith Predictor (Green Line: Controller 3)